Sine rule (Extended)

Core Concepts: Sine rule (Extended)

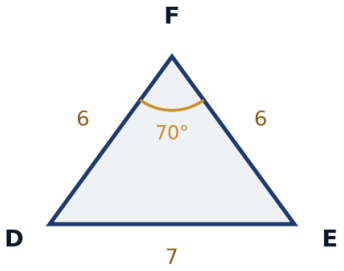
1 For the model $f(x) = 6x + 8$ in sine rule (extended), evaluate $f(2)$.

a. 20

b. 26

c. 12

d. 24



2 Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

3 A data set recorded for sine rule (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Procedural Fluency & Algebraic Manipulation

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4 For the model $f(x) = 6x + 8$ in sine rule (extended), evaluate $f(3)$.

a. 26

b. 32

c. 18

d. 30

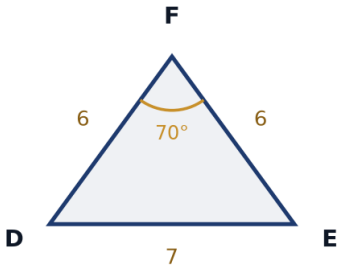
5 Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$



6 A data set recorded for sine rule (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

Graphical & Geometric Interpretation

7 For the model $f(x) = 6x + 8$ in sine rule (extended), evaluate $f(4)$.

a. 32

b. 38

c. 24

d. 36

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8 Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

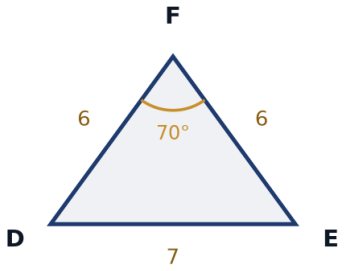
9 A data set recorded for sine rule (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5



Exam-Style Problem Solving & Modeling

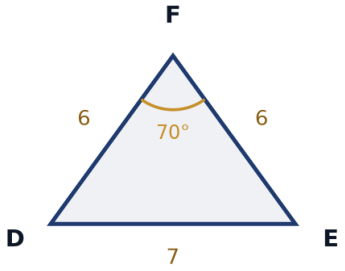
10 For the model $f(x) = 6x + 8$ in sine rule (extended), evaluate $f(5)$.

a. 38

b. 44

c. 30

d. 42



11 Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

a. $x = 4$ and $x = 10$

b. $x = 4$ and $x = 9$

c. $x = -4$ and $x = -9$

d. $x = 13$ and $x = 36$

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12 A data set recorded for sine rule (extended) consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5

d. 24.5

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